Unified Guidance Model:

ISO/PAS 8800, ISO 21448 (SOTIF) and the EU AI Act

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1 Introduction

This document describes the application and structure of the Unified Guidance Model (UGM), a harmonized verification and assurance framework integrating ISO/PAS 8800, ISO 21448 (SOTIF), and the EU AI Act for safety-relevant requirements.

2 Getting Started

The goal is to help organizations design and verify AI systems in physical products by aligning domain-specific safety standards with cross-industry legal obligations.

The UGM is intended as a practical complement to my thesis. It provides a reusable structure that allows engineers, auditors, and compliance teams to trace safety-related requirements through the entire AI safety lifecycle, from concept to post-market monitoring.

2.1 The SetUp Sheet

Before applying any sub-attribute model, the user must define the context of the AI system. The AI System Setup Sheet captures the essential parameters that activate the correct branches of the UGM. It ensures that each evaluation starts with a clearly defined scope and that the risk classification and domain context are consistently applied across all sub-attribute analyses.

The parameters include:

- Identify the domain: Whether the system operates in an automotive or non-automotive environment determines which standards and/or regulations are applicable.
- Determine the risk level: Classification under the EU AI Act as high-risk or non-high-risk activates or deactivates specific regulatory obligations.
- Clarify organizational roles: Defining whether the actor is a Provider or Deployer assigns accountability within the model.
- The Input Space (or Operational Design Domain, ODD) describes the range of conditions in which the AI system is intended to operate.

Together, these entries provide the contextual information that every GSN-based sub-attribute model requires to function correctly.

3 General Model Skeleton: Instructions to use the UGM

After completing the AI System Setup Sheet, the user proceeds to the verification phase. The UGM provides a reusable base structure for verifying each sub-attribute of AI system safety and trustworthiness.

3.1 Model Nature and Structure

The UGM is based on the Goal Structuring Notation (GSN), a standardized graphical method for expressing structured assurance arguments. GSN represents logical reasoning in a traceable and auditable form by connecting claims, assumptions, strategies, and evidence. In the context of this guidance, each GSN element is used to demonstrate that the AI system fulfills a specific sub-attribute derived from the extended Quality Management System. Each sub-attribute therefore becomes a self-contained assurance argument built from the same foundational skeleton, allowing consistency and reuse across different contexts, standards, and risk levels.

3.2 Node Types and Their Functions

Before applying the model, users must understand the standardized node types that structure every argument. The table below defines the base nodes that are used in every sub-attribute model.

Node	Definition
G1	Main goal: definition of the sub-attribute to be verified.
S1	Strategy: split according to Risk Level and Domain (Autonomous vs. Non-Autonomous).
S2	Sub-strategy: High-Risk and Autonomous context active.
S3	Sub-strategy: Non-High-Risk and Autonomous context active.

Table 1: Base Nodes and their defintions

Node	Definition
S4	Sub-strategy: High-Risk and Non-Autonomous context active.
S5	Sub-strategy: Non-High-Risk and Non-Autonomous context active.
C1	Context: system classified as High-Risk under the EU AI Act.
C2	Context: system classified as Non-High-Risk.
C3	Context: system operates as or within an Autonomous Vehicle.
C4	Context: system operates in a Non-Autonomous domain.
C5	Context: GPAI condition; activated only if system qualifies as General-Purpose AI.
SComp	Strategy: group complementary requirements from different standards addressing the same concept.
GNoREQ	Goal: indicates absence of explicit normative requirements for this sub-attribute in the given context.
JNoREQ	Justification: explains that no mapped requirements exist for this sub-attribute.
A1	Assumption: prerequisites and supporting information from ISO/PAS 8800 and general chapters of SOTIF are considered valid.

Table 1: Base Nodes and their defintions

These nodes define the invariant skeleton of the Unified Guidance Model. Context nodes (C1–C5) determine which branches are activated according to the setup sheet (Risk Level, Domain, GPAI). Strategy nodes (S1–SComp) define how requirements are decomposed or grouped, and Goal nodes (G1, GNoREQ) capture what must be demonstrated. Assumptions (A1) and Justifications (JNoREQ) provide traceability to prerequisites and rationale, while Solution nodes (Sn) represent tangible verification evidence.

3.3 Reusable Model Skeleton

The general structure is designed to be adaptable and repeatable. Every sub-attribute follows the same core logic:

- The main goal (G1) defines what sub-attribute is being verified.
- The first strategy (S1) splits the model according to risk and domain context.
- Within each branch, context nodes activate the relevant standards.
- The sub-claims represent the applicable normative requirements.
- These sub-claims are linked to solution nodes (Sn) that contain the concrete verification evidence.

This modularity allows the same skeleton to be applied to any sub-attribute by substituting the relevant requirements and evidence. It also supports the combination of complementary or overlapping regulatory obligations in a single argument tree.

3.4 Structural Variants for Requirements Relationships

Depending on how requirements from the different sources relate to each other (see Annex E of my thesis), four structural configurations exist within the model:

Relationship Type	Description	Structure
Direct Overlap	Two requirements cover the same content. Represented as one shared sub-goal with a justification node explaining the overlap and a single solution node.	Single branch
Partial Overlap	One requirement is broader or more detailed than the other. Represented as a main sub-goal and a subordinate goal for the narrower requirement. Both lead to a single solution node, justified by a “Partial Overlap” node.	Nested branch
Complementary	Requirements address different aspects of the same concept. Represented under a shared *SComp* strategy node, with one sub-goal for each requirement and separate solution nodes.	Parallel branch
No Overlap	Requirements relate to distinct concepts. Represented as two separate sub-goals under the same strategy node, each with its own solution node.	Separate branches

Table 2: Types of Relationships Between Requirements

These four patterns form the basis for representing every type of relationship between the EU AI Act, ISO 8800, and SOTIF requirements. They can be extended or nested as necessary to form complex, multi-level assurance trees.

3.5 Overview of the Skeleton

The figure below shows the base structure of the UGM, which can be adapted to any sub-attribute. The model is intended to illustrate traceability between risk

context, domain, requirements, and evidence.

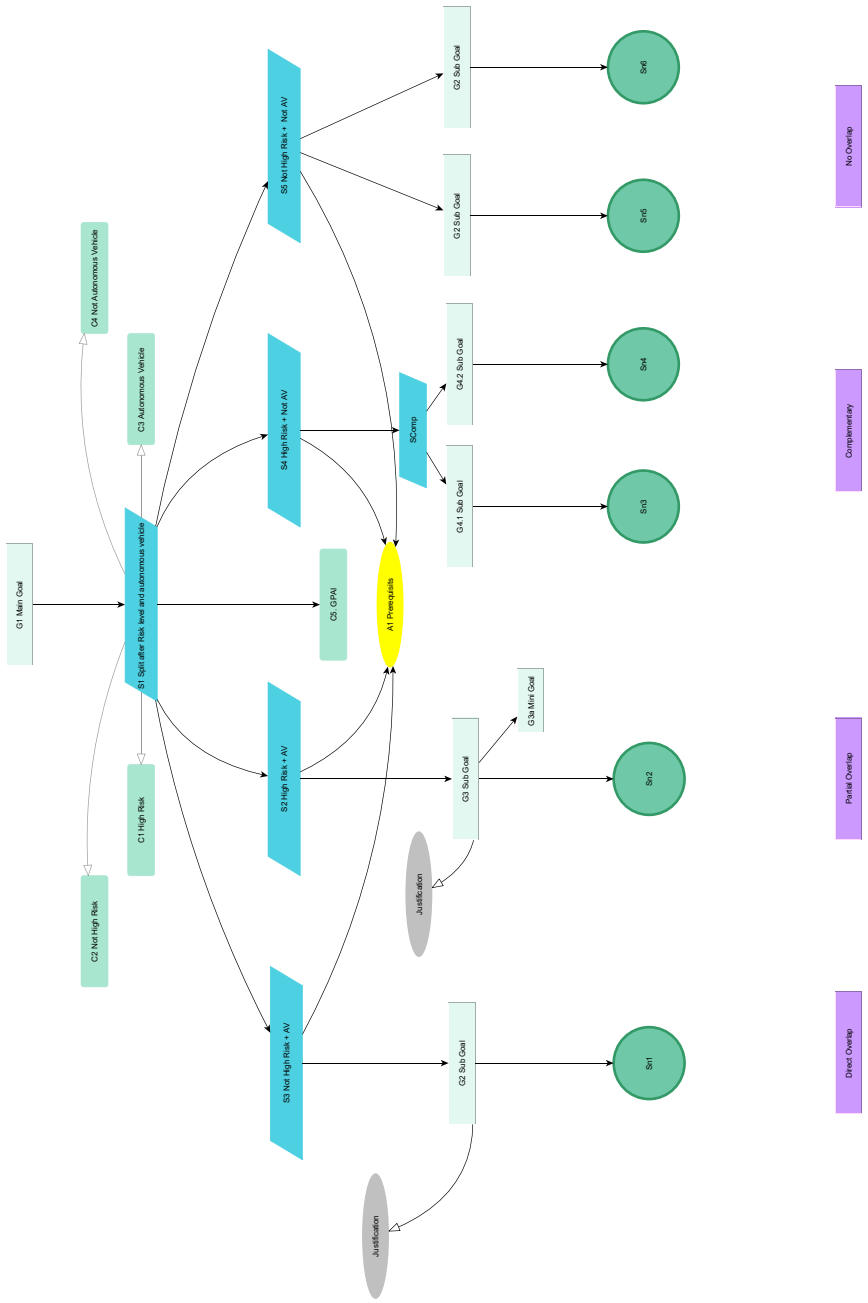

Figure 1: Overview of the reusable GSN skeleton

3.6 Practical Use

To apply this model:

- Start with the AI System Setup Sheet.
- Identify which context nodes (C1–C5) are active.
- Select the relevant sub-attribute schema.
- Determine the applicable requirement relationship type (direct, partial, complementary, no overlap).
- Populate the GSN skeleton accordingly.
- Attach evidence (Sn) for verification.

This structure ensures that the Unified Guidance Model is not a static diagram but a functional template, adaptable to any AI system, domain, or regulatory combination, while maintaining methodological consistency and traceability.

4 Example: Functional Completeness

This section illustrates how the Unified Guidance Model (UGM) is applied to a single sub-attribute. The example demonstrates the instantiation of the introduced reusable skeleton, filled with the specific requirements from ISO/PAS 8800, ISO 21448 (SOTIF), and the EU AI Act.
Provider: PPR
Deployer: DPY

Node	Definition	source	Role	
G1	Capability of a product to provide a set of functions that covers all the specified tasks and intended users' objectives	ISO/IEC 25010:2023		
C6	The system is in the safety lifecycle phase: requirement specification / verification / validation.	9.3.1 ISO/PAS 8800	PPR	
G2.1	A verification and validation strategy incl. validation targets and scenario space coverage is defined.	9.1.a (2) ISO 21448	PPR	
G2.2	The input space of the AI system is defined sufficiently to initiate the AI safety lifecycle.	9.3.1 ISO/PAS 8800	PPR	
G3.1	A justification is provided that refined AI safety requirements are reasonable to meet allocated safety requirements or mitigate functional insufficiencies.	9.3.3 ISO/PAS 8800	PPR	
G3.2	System-level argument for SOTIF release decision.	12.1c ISO 21448	PPR DPY	+
G4	Work products resulting from SOTIF activities are reviewed for completeness, correctness, and consistency.	12.1a ISO 21448	PPR	
G4a	Cases of non-conformance or input space limitations are identified and reported to the system development process.	9.3.5 ISO/PAS 8800	PPR DPY	+
G5.1	Test cases adequately verify the AI safety requirements of the AI component within the specified input space.	12.3.5 ISO/PAS 8800	PPR	

Table 3: Node Description of Functional Completeness

Node	Definition	source	Role	
G5.2	A verification and validation strategy for SOTIF, including validation targets, is defined.	9.1 a SOTIF	PPR	
G6	A verification and validation strategy for SOTIF, including validation targets, is defined.	9 (6) AIA	PPR	
Sn1	Input space definition (refined).	9.6.1 ISO/PAS 8800	PPR	
Sn2	Documented justification report that refined AI safety requirements are adequate.	9.6.2 ISO/PAS 8800	PPR	
Sn3	Report listing non-conformance cases and partial input space coverage cases.	9.6.3 ISO/PAS 8800	PPR	
Sn4	AI system verification report	12.6.1 ISO/PAS 8800	PPR	
Sn5	Definition of the verification and validation strategy	9.4 ISO 21448	PPR	
Sn6	SOTIF release argument	12.5 ISO 21448	PPR DPY	+
Sn7	Continous test documentation demonstrating that the high-risk AI system has been tested.	Recital 65 AIA	PPR	
J1	ISO/PAS 8800 (9.3.1) and SOTIF (9.1a/2) both define system boundaries (Input Space vs Scenario Space).	12.6.1 ISO/PAS 8800 & 9.1a/2 ISO 21448	PPR	
J2	ISO/PAS 8800 provides justification at requirement level; SOTIF requires argumentation at system level. Together they form a comprehensive release decision rationale.	9.3.3 ISO/PAS 8800	PPR DPY	+

Table 3: Node Description of Functional Completeness

Node	Definition	source	Role	
J3	The non-conformance report (8800) is a specific input that must be included in the SOTIF review (12.1a). 8800 ¡ SOTIF, because SOTIF requires a comprehensive review of all work products, while 8800 mandates specific reporting.	9.3.5 ISO/PAS 8800 & 12.1a SOTIF	PPR DPY	+
J4	ISO/PAS 8800 Clause 12.3.5 specifies the design and adequacy of test cases, which operationalizes and concretizes the overarching SOTIF requirement for a defined V&V strategy (Clause 9.1 a).	12.3.5 ISO/PAS 8800 & 9.1 a SOTIF	PPR	
J5	Art. 9(6) requires testing of high-risk AI systems to ensure consistent performance and compliance. To demonstrate this, complementary requirements from G5.1 8800 and G2.1 SOTIF are considered.	12.3.5 ISO/PAS 8800 & 9.1 a SOTIF & 9(6) AIA	PPR	
J6	Only G5.1 / 12.3.5 ISO/PAS is complementary with AIA Art. 9(6). Other 8800 requirements are additional but not directly paired	12.3.5 ISO/PAS 8800 & 9 (6) AIA	PPR	

Table 3: Node Description of Functional Completeness

The Functional Completeness model follows the complementary relationship type described in Section 3.4. A central SComp strategy node aggregates complementary and partially overlapping requirements from ISO/PAS 8800, ISO 21448, and the EU AI Act. Together, these requirements establish a unified argument that ensure complete and reliable coverage of the AI system's defined input space.

The argument begins with Context C6, which anchors the model in the safety lifecycle phase of requirement specification, verification, and validation. This context activates the relevant SOTIF and ISO 8800 clauses. From there, G2.1 and G2.2 define the system-level setup. G2.1 ensures that the verification and validation strategy includes coverage of the relevant scenario space. G2.2 complements this by requiring a sufficiently refined input-space definition to initiate the AI safety lifecycle. Their relationship is justified in J1, which links scenario-space and input-space coverage as equivalent boundary-defining concepts.

Subsequent goals address requirement refinement and validation reasoning. G3.1 and G3.2 combine ISO 8800 and SOTIF requirements, integrating requirement-level justification with system-level release argumentation. J2 documents this complementarity.

The verification and testing layer includes G5.1, G5.2, and G6, which collectively demonstrate compliance across all three frameworks. Evidence nodes (Sn1–Sn7) provide the concrete verification outputs supporting these claims. The aggregated structure demonstrates that functional completeness is interpreted as a full and justified coverage of the defined input or scenario space and is verified through harmonized evidence across all three regulatory sources. The complementary integration of ISO 8800, SOTIF, and AI Act requirements ensures that both domain-specific safety and cross-industry compliance are met within a single assurance argument.

Figure 2: Legend of node and role representations used in the Unified Guidance Model (UGM). Node types (G, C, S, Sn, J, A) are distinguished by shape, while border colors denote responsible roles: green for Provider (PPR), blue for Deployer (DPY), and no border for shared responsibility.

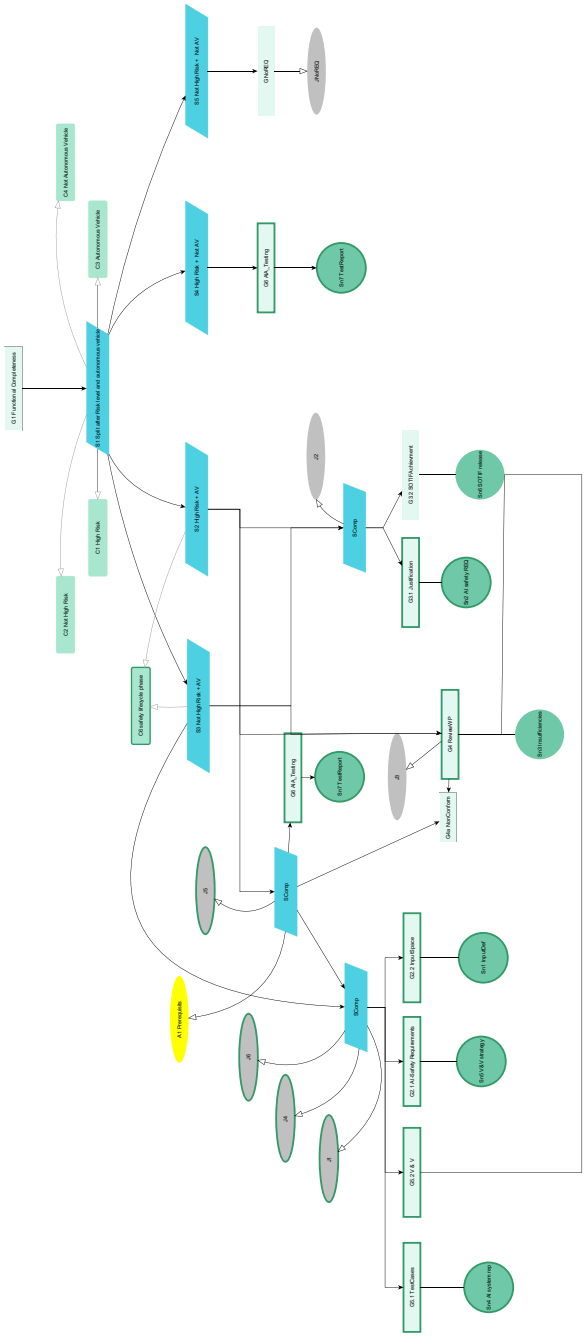

Figure 3: Overview of the application of the UGM on Functional Completness

5 Example: Fail Safe

Node	Definition	source	Role
G1	Capability of a product to automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure.	ISO/IEC 25010:2023	
G2	Adequate mitigation and control measures addressing risks that cannot be eliminated.	9 (5) b AIA	PPR
Sn1	Work product is the specification of SOTIF measures	8.5 ISO 21448	PPR
Sn2	Risk-management system	Recital 65 AIA	PPR
J1	SOTIF precisely defines a specific type of “control mea sure” that is required in the AIA.	8.1a ISO 21448 & Art. 9 (5) b AIA	PPR

Table 4: Node Description of Fail Safe

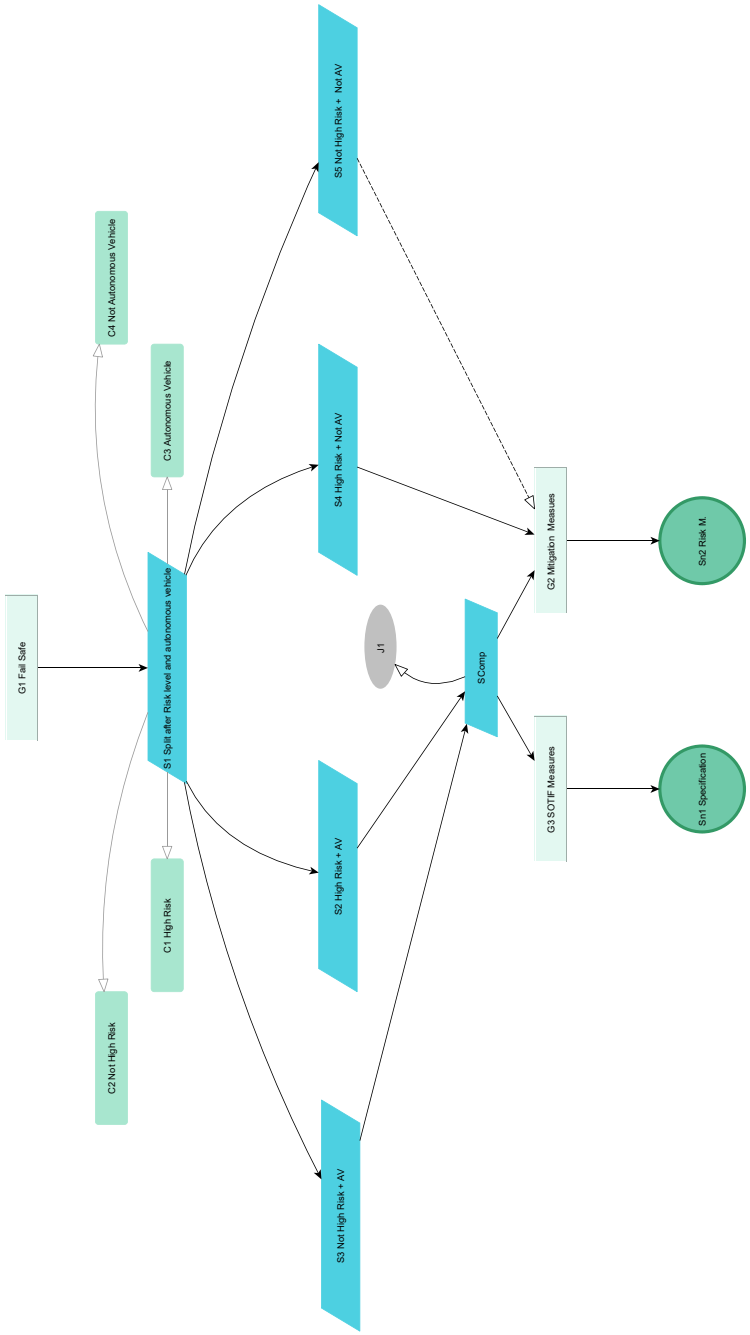

Figure 4: Overview of the application of the UGM on Fail Safe